

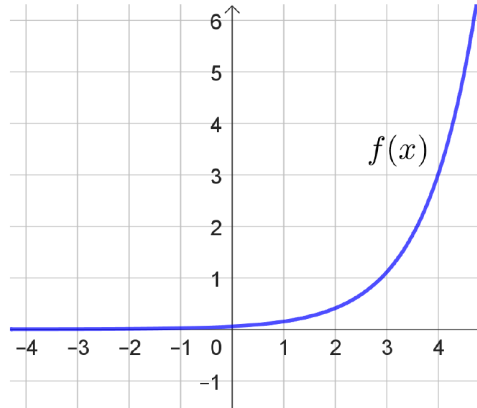
Sec 3.6 – The Chain Rule and Inverse Functions

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Recall, an inverse function is one where we switch inputs and outputs: if $f(a) = b$, then $f^{-1}(b) = a$.

Two different functions are given below. In each case, find $f^{-1}(3)$.

x	$f(x)$
0	7
1	5
2	3
4	1
5	-1

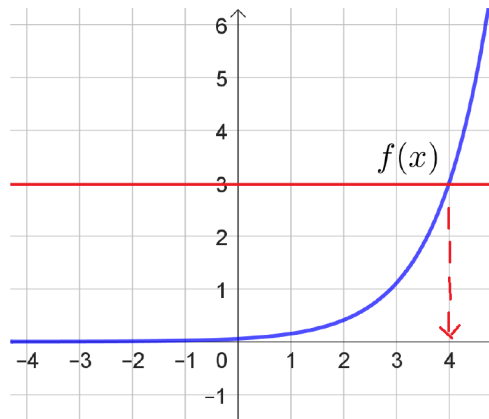


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We read backward from the y -value of 3 to get the x -value of 2: $f^{-1}(3) = 2$.



We find where the graph intersects $y = 3$ and read off the x -value of 4: $f^{-1}(3) = 4$.

How can we check/see two functions are inverses?

- The graphs are reflections over the line $y = x$
- We use the inverse to undo operations when solving equations (e.g. when you have an equation with a square, you undo it by using the inverse and taking the square root)
- $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$

The last suggests that chain rule might be involved with the derivatives...

We know that $\frac{d}{dx}x^{1/2} = (1/2)x^{-1/2}$. Let's find this another way.
 Since $g(x) = x^{1/2} = \sqrt{x}$ is the inverse function of $f(x) = x^2$ (for $x > 0$), we can write

$$f(g(x)) = (\sqrt{x})^2 = x$$

Take the derivative of both sides:

$$\begin{aligned} \frac{d}{dx}(g(x))^2 &= \frac{d}{dx}x && \text{by the Chain Rule} \\ 2g(x)g'(x) &= 1 && \text{by dividing by } 2g(x) \\ g'(x) &= \frac{1}{2g(x)} \\ g'(x) &= \frac{1}{2\sqrt{x}} = \frac{1}{2}x^{-1/2} && \text{by substituting } g(x) = \sqrt{x} \end{aligned}$$

The same can be done to show $\frac{d}{dx}x^{1/n} = \frac{1}{n}x^{1/n-1}$.

The functions $\ln x$ and e^x are inverses, so we can use the same method to determine the derivative of $\ln x$.

$$\begin{aligned} e^{\ln x} &= x \\ \frac{d}{dx}(e^{\ln x}) &= \frac{d}{dx}x \\ e^{\ln x} \frac{d}{dx}(\ln x) &= 1 \\ \frac{d}{dx}(\ln x) &= \frac{1}{e^{\ln x}} \\ \frac{d}{dx}(\ln x) &= \frac{1}{x} \end{aligned}$$

Function	Input	Output	Inverse	Input	Output
$\sin x$	$(-\infty, \infty)$ angle	$[-1, 1]$ number	$\arcsin x$ $(\sin^{-1} x)$	$[-1, 1]$ number	$[-\pi/2, \pi/2]$ angle
$\cos x$	$(-\infty, \infty)$ angle	$[-1, 1]$ number	$\arccos x$ $(\cos^{-1} x)$	$[-1, 1]$ number	$[0, \pi]$ angle
$\tan x$	$(-\infty, \infty)$ angle	$(-\infty, \infty)$ number	$\arctan x$ $(\tan^{-1} x)$	$(-\infty, \infty)$ number	$(-\pi/2, \pi/2)$ angle

We know $\tan(\arctan x) = x$. Taking the derivative of both sides:

$$\frac{d}{dx} \tan(\arctan x) = \frac{d}{dx} x$$

$$\frac{1}{\cos^2(\arctan x)} \frac{d}{dx} (\arctan x) = 1$$

We can simplify this derivative, using the trig identity

$$1 + \tan^2(\theta) = \sec^2(\theta) = \frac{1}{\cos^2(\theta)}.$$

Substituting $\theta = \arctan x$, we get

$$\frac{1}{\cos^2(\arctan x)} = 1 + \tan^2(\arctan x) = 1 + x^2$$

Therefore,

$$\frac{d}{dx} (\arctan x) = \frac{1}{1 + x^2}$$

We know $\sin(\arcsin x) = x$. Taking derivatives of both sides,

$$\frac{d}{dx} \sin(\arcsin x) = \frac{d}{dx} x$$

$$\cos(\arcsin x) \frac{d}{dx} (\arcsin x) = 1$$

$$\frac{d}{dx} (\arcsin x) = \frac{1}{\cos(\arcsin x)}$$

Since $\arcsin x$ is the angle with opposite leg x and hypotenuse 1, the Pythagorean theorem says that the adjacent leg a satisfies $x^2 + a^2 = 1$ or $a = \sqrt{1 - x^2}$. Using this triangle, we can see that the cosine of the angle is $\sqrt{1 - x^2}$. This is exactly $\cos(\arcsin x)$!

Therefore,

$$\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1 - x^2}}$$

Theorem

$$\frac{d}{dx} (\ln x) = \frac{1}{x} \quad \frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1 - x^2}} \quad \frac{d}{dx} (\arctan x) = \frac{1}{1 + x^2}$$

Partner up

Find derivatives of the following functions:

1. $f(x) = 3 \arcsin x + 2 \ln x - 5 \arctan x$

2. $f(x) = \frac{\arcsin x}{x^3}$

3. $f(x) = \ln(x^3 - 4x)$

4. $f(x) = 3^x \arcsin x$

5. $f(x) = \arctan(x^5)$

① $f(x) = 3 \arcsin x + 2 \ln x - 5 \arctan x$

$$f'(x) = \frac{3}{\sqrt{1 - x^2}} + \frac{2}{x} - \frac{5}{1 + x^2}$$

② $f(x) = \frac{\arcsin x}{x^3}$

$$f'(x) = \frac{(\arcsin x)' x^3 - \arcsin x (x^3)'}{(x^3)^2} = \frac{\frac{x^3}{\sqrt{1 - x^2}} - \arcsin x (3x^2)}{x^6}$$

③ $f(x) = \ln(x^3 - 4x)$

$$f'(x) = \frac{1}{x^3 - 4x} (x^3 - 4x)' = \frac{3x^2 - 4}{x^3 - 4x}$$

④ $f(x) = 3^x \arcsin x$

$$f'(x) = (3^x)' \arcsin x + 3^x (\arcsin x)' = (\ln 3) 3^x \arcsin x + 3^x \frac{1}{\sqrt{1 - x^2}}$$

⑤ $f(x) = \arctan(x^5)$

$$f'(x) = \frac{1}{1 + (x^5)^2} (x^5)' = \frac{5x^4}{1 + x^{10}}$$

Theorem

$$\frac{d}{dx}(f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}$$

Here are values for f and f' :

x	$f(x)$	$f'(x)$
0	-1	3
1	2	2
2	4	1
3	5	0

Find

(a) $f(2)$

(b) $f^{-1}(2)$

(c) $f'(2)$

(d) $(f^{-1})'(2)$

(e) $(f^{-1})'(5)$

a $f(2) = 4$

b $f^{-1}(2) = 1$

c $f'(2) = 1$

d $(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(1)} = \frac{1}{2}$

e $(f^{-1})'(5) = \frac{1}{f'(f^{-1}(5))} = \frac{1}{f'(3)} = \frac{1}{0}$ DNE! This would be a vertical tangent line